Final Capstone Project Description for Group 9226

Data Analytics for AI Business
Programme 3_MBAE_S25_MK_AIB_A

Graduate School of Business (GSB), Miami
Arab Academy for Science, Technology and Maritime Transport (AASTMT)

Executive Business Analytics and AI Enabled Decision Support

A Portfolio of Enterprise Decision Intelligence Projects

1 Overview

The final capstone is an integrated assessment of business analytics, business intelligence, Artificial Intelligence and executive communication. Each team acts as an executive analytics advisory group, using its assigned enterprise dataset to investigate a business problem and develop recommendations that a management audience can assess.

The project follows a decision-focused approach. Students should begin with the decisions and business questions, establish trustworthy measures, examine performance and evaluate suitable AI applications. Analytical complexity is useful only when it helps answer those questions and the team can explain its results and limitations.

Group 9226 comprises five thematic project teams. The portfolio covers enterprise financial performance, quality in asset-intensive projects, predictive maintenance, resource allocation and digital transformation. Each team applies the same analytical principles while preserving the operating and financial definitions of its assigned scenario.

The supplied enterprise datasets are fully synthetic educational panels with embedded business rules, operational relationships and derived indicators. Their formulas and simulated patterns support analytical exercises; they are not observed company outcomes. Correlation alone does not establish causation, and predictive performance does not demonstrate achieved savings or operational improvement.

Students will produce a working Power BI dashboard, an executive presentation and a comprehensive analytical report. The supplied reference publication explains the case and dataset; the submitted capstone must contain the team's own validation, analysis, implemented dashboard, model evaluation and recommendations.

2 Learning Outcomes

On successful completion, students should be able to demonstrate the following capabilities across the enterprise analytics lifecycle:

• Translate a business scenario into explicit executive questions and decision objectives.

• Interpret the assigned dataset, its grain, dimensions, information timing and synthetic assumptions.

• Validate completeness, uniqueness, business rules, accounting identities and KPI formulas.

• Conduct systematic exploratory data analysis of distributions, trends, relationships and anomalies.

• Engineer and validate derived measures with correct denominators and financial scope.

• Build interactive Power BI dashboards with a coherent data model, navigation and drill-through.

• Connect descriptive and diagnostic analysis with suitable predictive or prescriptive applications.

• Select and evaluate AI methods against informative baselines using appropriate validation.

• Explain model performance, uncertainty and practical limits in business language.

• Develop recommendations that recognize operating constraints and the evidence needed for implementation.

• Communicate findings through a professional report and a concise executive briefing.

• Use AI tools responsibly, disclose their contribution and retain human responsibility for the final judgement.

Evidence of learning

The dashboard, report and presentation should tell the same business story. KPI definitions, totals, model metrics and recommendations must reconcile across all three deliverables. Every team member should be able to explain the decisions, evidence and limitations, including the parts developed by other members.

3 Project Structure and Team Allocation

The Miami cohort portfolio uses the following five project themes. The official allocation and each team's supplied package govern the detailed business setting and scope.

Team	Project theme	Executive decision domain
1	AI-Enabled Enterprise Performance and Financial Decision-Support System	Financial, commercial, operational and service performance
2	Predictive Quality and Operational Performance Analytics for Asset-Intensive Projects	Quality, inspections, rework, cost and schedule
3	AI-Driven Predictive Maintenance and Asset Reliability Decision-Support System	Condition, failures, work orders, downtime and economic impact
4	AI-Assisted Resource Allocation and Business Performance Optimization	Capital, workforce, equipment, inventory and service capacity
5	AI-Enabled Operational Intelligence and Digital Transformation Dashboard	Cost control, integration, digital adoption and operational reliability

Integrated team effort

Each team functions as an Executive Analytics Advisory Group. Individual submissions are not permitted. Team membership follows the official Group 9226 allocation. Each team should preserve the business scope and controlled definitions in its assigned package.

Assign clear responsibilities, maintain a common analytical file structure and record the work completed by each member. Technical tasks may be distributed, but the final dashboard, report and presentation must be integrated and consistent.

Dataset and reference materials

Use the assigned CSV, supporting analytics workbook and data dictionary as the analytical basis. The published project reference supports understanding of the scenario, variable definitions and potential dashboard designs. Teams should verify supplied results, develop their own analysis and clearly distinguish reference material from new findings.

Preserve the approved master dataset. Perform transformations in a documented analytical copy or Power Query workflow. If a material business-rule issue is found, document it and consult the instructors before changing the underlying scenario or silently rewriting source values.

4 Project Scope

4.1 Dataset Governance

Establish the analytical grain and data dictionary before calculating executive results. Check row and variable counts, missing values, duplicates, data types, dimensional consistency and date coverage. Validate operational conservation, inventory continuity and financial formulas where applicable.

- Record the checks performed and summarize passed checks and unresolved exceptions.
- Distinguish source-quality measures from missing values in the canonical dataset.
- Document zero-denominator handling, repeated counts and exclusions from financial measures.
- Retain a traceable transformation workflow and reconcile results to the assigned master.

The instructor's internal dataset engineering process is separate from the student submission. The student report should present its own governance methodology and evidence, not reproduce the internal engineering workflow as a publication section.

4.2 Exploratory Data Analysis

Investigate distributions, operational and financial trends, customer or account patterns where represented, correlations, outliers, anomalies and executive indicators. Use business interpretation to explain the results and their significance for the assigned decisions.

- Use appropriate descriptive statistics and disclose the relevant populations and periods.
- Review performance by enterprise dimensions and compare like-for-like units.
- Distinguish accounting or operational identities from independent associations.
- Investigate outliers before removal; preserve valid exceptional observations.
- Avoid interpreting fictional geography or simulated relationships as measured real-world evidence.

EDA provides the foundation for dashboard measures, feature selection and modelling. Include clear evidence of how exploratory findings informed later analytical choices.

4 Project Scope Continued

4.3 Data Transformation and Feature Engineering

Prepare the data for Power BI and any selected AI method. Set appropriate data types, construct dimensions, document transformations and validate engineered variables. Derive executive KPIs with explicit formulas, business meanings and portfolio aggregation rules.

Use matching summed numerators and denominators where appropriate. Do not average row rates without considering their weights, double-count linked indicators or sum repeated entity counts as unique customers or employees. Distinguish predictors available before a decision from outcomes observed afterward.

4.4 Executive Dashboard Development

Build executive-ready Power BI views with KPI scorecards, interactive slicers, navigation, drill-through, trends and comparative analysis. Geographic views are appropriate only where meaningful locations are provided. Use consistent currency, units, labels and visual formatting.

The project reference contains five suggested dashboard designs. Adapt and implement suitable views within a coherent PBIX file. The requirement for at least five visualization types is distinct from the number of dashboard pages; choose page count and layout to support the assigned business decisions.

4.5 Artificial Intelligence Applications

Investigate suitable classification, clustering, regression, forecasting, recommendation or prioritization methods. Select methods because they answer a business question; every listed technique need not be applied. Each selected model requires business justification, transparent inputs, evaluation and executive interpretation.

Compare predictive models with an informative baseline. For time-dependent scenarios, use chronological validation and prevent current-outcome leakage. Report relevant metrics, error patterns and limitations. Clustering requires a clear profile interpretation and a stability or suitability assessment. State whether an application was executed or only proposed.

4.6 Executive Recommendations

Recommendations should address the identified problems, cite analytical evidence, recognize constraints and identify a practical owner or next step. Separate demonstrated findings from proposed actions and unvalidated impact estimates. Optimization claims require explicit objectives, feasible constraints and a solved comparison; descriptive rankings alone do not establish an optimal decision.

5 Mandatory Project Deliverables

Each team must submit three integrated deliverables. Use the formats below and keep all claims and measures consistent across files.

A Enterprise Power BI Dashboard

Required file format: .pbix

- An executive landing page explaining the key decision and portfolio performance.
- A minimum of five appropriate visualization types across the report.
- Executive KPIs with validated definitions and aggregation.
- Interactive slicers and navigation that support coherent exploration.
- Drill-through into relevant entities or operational exceptions.
- Trend and comparative views with clear units and reporting periods.
- Professional formatting and decision-oriented visual storytelling.

The PBIX must open and function as submitted. Check filters, relationships, measure results and drill-through interactions. Where a local CSV source is used, load the data into the model and document the source and refresh procedure in the PDF report; do not rely on an unavailable personal path for the reviewer to inspect the saved dashboard.

B Executive Presentation

Required file format: .pptx
Recommended length: 12-15 professional slides

- Business context, executive problem and key questions.
- Dataset scope and a concise account of governance and methodology.
- Demonstration of the implemented dashboard and its decision value.
- Selected AI application, evaluation against a baseline and limitations.
- Numerical executive insights and evidence-based recommendations.
- Decision implications, proposed next steps and responsibility.

The presentation should prioritize a coherent executive story. Detailed calculations and supporting diagnostics belong in the report, with only the evidence needed for the management discussion shown on slides.

5 Mandatory Project Deliverables Continued

C Comprehensive Analytical Report

Required file format: .pdf
Recommended length: 20-25 pages

The report should document the team's completed analytical work. The supplied reference publication is a case resource, not a replacement for this submission. Its variable matrices and suggested designs may be summarized or referenced, while the capstone report presents actual governance, EDA, dashboard implementation, model results and recommendations.

Report component	Expected evidence
Executive Summary	Business problem, main findings, decisions and limits
Introduction and Business Context	Enterprise setting, stakeholders and scope
Business Questions	Explicit questions aligned with the assigned decisions
Dataset Description	Grain, coverage, dimensions, variables and synthetic provenance
Methodology and Governance	Checks, transformations, reconciliation and unresolved issues
Exploratory Data Analysis	Distributions, trends, relationships, anomalies and interpretation
Dashboard Design and Implementation	Actual views, measures, interactions and decision use
AI Applications	Inputs, timing, evaluation, baseline and model limitations
Executive Insights	Numerical findings traceable to the verified analytical data
Strategic Recommendations	Evidence, constraints, priorities and practical next steps
Conclusion	Decision implications and further evidence required
AI Usage Disclosure	Tools, purposes, assistance scope and human validation

Analytical traceability

Show formulas, relevant denominators and model metrics with enough detail for review. Place supplementary validation tables and modelling evidence within report appendices as needed. Keep references and team contribution information within the same PDF so the submission stays within the three-file limit.

6 Executive Presentation Requirements

Each team will deliver an executive briefing of approximately 15 minutes. The session should simulate a meeting with senior management or a board. Presentation scheduling and any separate question period will be announced by the instructors.

Executive briefing expectations

- Define the management problem and the decision your analysis supports.
- Demonstrate the dashboard and explain how its views answer the business questions.
- Present the strongest numerical findings with clear business meaning.
- Explain the selected AI application, baseline comparison and performance limits.
- Prioritize practical recommendations and disclose constraints and unvalidated benefits.
- Respond to questions about data definitions, measures, model choices and team judgement.
- Manage time and coordinate transitions between presenters.

Professional communication

Assessment considers executive presence, clarity, confidence, storytelling, business relevance, evidence-based recommendations, time management and response to questions. Technical detail should be included where it supports a management decision, with terminology explained for the intended audience.

Suggested briefing sequence

Part	Focus
Opening	Business context and decision objectives
Evidence	Validated KPIs, EDA findings and dashboard demonstration
AI evaluation	Selected method, baseline, results and limits
Recommendation	Priorities, constraints, next steps and decision request

Rehearse as a team and verify that slides, dashboard and report use the same totals and definitions. Every member remains responsible for understanding the integrated submission. Report a weak model honestly rather than replacing evidence with an unsupported claim of success.

7 Assessment Criteria

Total assessment: 90 marks

Assessment component	Marks	Assessment focus
Dataset Governance and Data Preparation	10	Validation, consistency, business-rule checks, transformations and KPI integrity.
Exploratory Data Analysis	10	Patterns, trends, anomalies, relationships and sound business interpretation.
Executive Power BI Dashboard Design	15	Executive relevance, architecture, KPIs, interactivity, navigation, drill-through and usability.
Analytical Depth and Executive Insights	15	Integration of variable types, numerical evidence, interpretation and decision value.
AI Applications and Strategic Recommendations	10	Method suitability, evaluation, baseline comparison, explainability and practical recommendations.
Comprehensive Written Report	10	Methodology, organization, coherence, analytical rigor and professional communication.
Executive Presentation and Professional Communication	20	Executive presence, clarity, storytelling, demonstration, time management and responses.
Total	90	Integrated enterprise analytics and executive decision support

Evaluation emphasizes analytical quality, business understanding, defensible conclusions and professional communication. A model that does not outperform its baseline can still demonstrate strong analytical judgement when its evaluation is rigorous and its limitations are communicated.

8 Submission Requirements

Upload one team package through Moodle containing exactly three files: the Power BI dashboard (.pbix), executive presentation (.pptx) and comprehensive report (.pdf). Individual submissions are not permitted.

Use the naming pattern Group_9226_Team_N_Dashboard.pbix, Group_9226_Team_N_Presentation.pptx and Group_9226_Team_N_Report.pdf, replacing N with your team number. Confirm that all files open correctly and that the PBIX contains its saved analytical data.

Submission deadline: To be announced. The instructors will confirm the deadline through Moodle. Late submissions require official approval in accordance with Graduate School regulations. Include supporting analyses and disclosure within the required files; do not add a fourth upload.

9 Responsible AI and AI Usage Disclosure

AI tools may support brainstorming, coding, documentation, visualization and drafting. Each team remains responsible for interpreting the dataset, selecting models, validating calculations, exercising business judgement and producing its final recommendations and conclusions.

Required disclosure

Include an AI Usage Disclosure Statement in the report that specifies:

- The AI tools used.
- The purpose of each use and the scope of assistance.
- The tasks and content generated or revised with AI support.
- The checks performed by team members to validate calculations, code, sources and conclusions.
- Confirmation that final analytical judgement and responsibility remain with the project team.

Distinguish a predictive or analytical model from a generative AI assistant. Explain both where used. Do not present AI-generated text, invented figures or unverified model results as analytical evidence. Preserve the synthetic provenance of the assigned data and acknowledge the limits of generalization.

10 Final Guidance

The capstone demonstrates how a management team can turn a defined business problem and trustworthy analytical data into an executive discussion. Strong work combines validated measures, appropriate analysis, a usable dashboard, honest model evaluation and recommendations that respect the evidence.

Use the assigned enterprise framework consistently, but adapt the interpretation to the scenario. Explain what happened in the synthetic setting, what management should investigate and what additional information is needed before acting. A descriptive association is not a proven causal effect, a forecast is not an optimized plan, and proposed benefits are not realized gains.

Maintain an integrated team workflow, review all deliverables together and prepare to defend the reasoning behind the conclusions. Every visual should answer a business question, every selected model should support a decision and every recommendation should state its evidence and practical limits.

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